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On the Nomenclature of the Peroxides and the Peroxides of the Aldehydes.

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Nomenclature of the Peroxides

Now that an overview of the most important forms of the peroxides has been gained, it is time to make a proposal for the nomenclature of these compounds, since their naming has hitherto not followed any uniform principle.

In doing so, we adhere as far as possible to the customary nomenclature, but shorten the designation "hydrogen peroxide" (*Wasserstoffsuperoxyd*) by calling it "**hydroperoxide**." The names "persulfuric acid" (*Perschwefelsäure*) and "persulfate" are retained as established; benzoyl hydrogen peroxide receives the name "**benzopersulfuric acid**" (*Benzopersäure*).

The hydroperoxides correspond to the "hydroxides," the "peroxides" to the "oxides." With this nomenclature one encounters the difficulty that the class name for persulfuric acid and for Caro's acid becomes the same, namely "**peracid**" (*Persäure*); we therefore propose to regard the designation "persulfuric acid" (*Perschwefelsäure*) as an abbreviation for "peroxide-sulfuric acid" (*Peroxydschwefelsäure*), and accordingly to say: the acid of potassium persulfate belongs to the class of the **peroxide acids** (*Peroxydsäuren*), that of Caro's reagent to that of the **peracids** (*Persäuren*).

We further report that phthalmonoperacid and peroxyphthalic acid have been prepared by us and are to be described more closely in due course. A name for the simple peroxide of acetone we have not included in the table, because we have not succeeded in preparing a substance of this class. One may, incidentally, designate it, in agreement with the other names, as "acetone peroxide."

We cannot, moreover, close these introductory remarks without mentioning that Mr. Bodländer[1)], as we only subsequently saw, had predicted in hypothetical fashion the existence of benzoperacid and its role in the autoxidation of benzaldehyde on the basis of earlier experiments, and we are pleased that the facts have proved him right.

Formula	Old name	New name
H_2O_2	Hydrogen peroxide	Hydroperoxide
$C_6H_5CO\cdot O_2H$	Benzoyl hydrogen peroxide	Benzopersulfuric acid
$C_6H_5CO\cdot O_2\cdot CO\cdot C_6H_5$	Benzoyl peroxide	Benzoperoxide
$HO\cdot SO_2\cdot O_2H$	Caro's acid	Sulfomonoperacid
$HO\cdot SO_2\cdot O_2\cdot SO_2\cdot OH$	Persulfuric acid	Persulfuric acid
$HO\cdot CO\cdot C_6H_4\cdot CO\cdot O_2H$	—	Phthalmonoperacid
$HOOC\cdot C_6H_4\cdot CO\cdot O_2\cdot CO\cdot C_6H_4\cdot COOH$	—	Peroxyphthalic acid
$CCl_3\cdot CH(OH)\cdot O_2H$	—	Chloral hydroperoxide
$CCl_3\cdot CH(OH)\cdot O_2\cdot (OH)CH\cdot CCl_3$	—	Dichloral peroxide hydrate
$C_6H_5\cdot CH(OH)\cdot O_2\cdot (OH)CH\cdot C_6H_5$		Dibenzal peroxide hydrate

Formula	Old name	New name
	Diphenylformal peroxide hydrate (Nef)	
$C_6H_5{\cdot}CH<O_2/O_2>CH{\cdot}C_6H_5$	—	Dibenzal diperoxide
$CH_2(OH){\cdot}O_2{\cdot}CH_2(OH)$	Formal peroxide hydrate	Diformal peroxide hydrate
$(CH_3)_2C<O_2/O_2>C(CH_3)_2$	Polymeric acetone peroxide	Diacetone diperoxide
$[(CH_3)_2C{\cdot}O_2]_3$	Tricycloacetone peroxide (Wolffenstein)	Triacetone triperoxide

Note: in the two-O_2 bracket formulas above, "$<O_2/O_2>$" denotes a cyclic diperoxide bridge (two –O–O– links joining the two carbons).

The Peroxides of the Aldehydes

Of peroxides of the aldehydes only two were hitherto known, a formaldehyde compound of Legler[2)] and a benzaldehyde derivative of Nef[3)].

We have found that the aldehydes behave toward hydrogen peroxide and toward Caro's reagent similarly to the ketones. Hydrogen peroxide, as Nef already observed, shows no particular tendency to oxidize aldehydes; sulfomonoperacid (Caro's acid) gives with aldehydes and ketones, depending on the conditions, peroxides or oxidation products.

Peroxides of Chloral.
Dichloral Peroxide Hydrate.

This compound forms directly from chloral and hydrogen peroxide.

To an ethereal solution of hydrogen peroxide, which had been prepared by extracting a concentrated solution with ether and drying with sodium sulfate, a small excess of chloral was added. After evaporation of the ether in vacuum, a syrupy mass remained which, using 5 g chloral, solidified within 24 hours into a hard, crystalline mass.

The same substance is obtained more conveniently by use of Caro's acid.

5 g chloral were added to a cold mixture of 10 g potassium persulfate and 50 g sulfuric acid hydrate and left standing at ordinary temperature. The crystal separation that begins after a short time is complete after 3 hours. One then pours onto ice, takes up the substance with ether, dries it, and drives off the ether in vacuum. To remove chloral hydrate, the mass was triturated with water and dried on clay. Purification was carried out by precipitating the solution in as little ether as possible with benzine (ligroin) and recrystallizing from warm benzene. This caused difficulties, since only after many attempts was a substance of definite composition obtained.

From benzene or chloroform the substance crystallizes in six-sided leaflets or plates of m.p. 122°. On melting, decomposition occurs simultaneously with strong gas evolution, which is why the melting point is found to differ by a few degrees according to the manner of heating. The substance yields a crystalline ether compound. When it is poured over, in the dry state, with ether, it bakes together under slight warming into a crystal mass that dissolves readily in more ether. On spontaneous evaporation the ether compound crystallizes out in long prisms, which weather very rapidly in air with loss of ether. The analysis gave an ether content of one molecule.

2.5682 g substance lost on standing in vacuum, to constant weight, 0.4427 g in weight.

$C_4H_4Cl_6O_4 + C_4H_{10}O$. Loss: Calc. 18.38. Found 17.24.

The analysis of the substance recrystallized from benzene gave the following values:

0.3168 g substance: 0.1771 g CO_2, 0.0411 g H_2O. — 0.2616 g substance: 0.6785 g AgCl.

$C_4H_4Cl_6O_4$. Calc. C 14.59, H 1.22, Cl 64.74.

Found C 15.25, H 1.44, Cl 64.16.

Although the carbon content was always found somewhat too high in this and many other analyses, it nevertheless follows from the molecular-weight determination given further below, as well as from the determination of the active oxygen, that no other formula can apply to the substance. From the analysis there follows, namely, the formula

$$CCl_3{\cdot}CH\overset{O\text{—}O}{\underset{OH\ \ HO}{}}CH{\cdot}CCl_3,$$

which corresponds to a content of 4.96 pct. active oxygen. To determine the latter, the substance was brought into solution by pouring over it a little sodium bicarbonate solution, and the hydrogen peroxide thereby formed was titrated in sulfuric-acid solution with permanganate, of which 1 cc corresponded exactly to one milligram of hydrogen peroxide.

0.1609 g substance: 16.6 cc permanganate solution.

$C_4H_4Cl_6O_4$. Calc. act. O 4.96. Found act. O 4.86.

In alcohol, ether, ethyl acetate, and glacial acetic acid the substance dissolves readily, less easily in benzene and chloroform. From the chloroform solution it is precipitated by benzine in six-sided plates, from the ethereal solution as the ether compound in needles.

If anhydrous chloral and anhydrous hydrogen peroxide are brought together in ethereal solution, there remains after evaporation of the ether, as stated above, a syrupy mass which could not indeed be examined, but which one may well assume with certainty to represent the addition compound corresponding to chloral hydrate, the **chloral hydroperoxide**, $CCl_3{\cdot}CH(OH)(O{\cdot}OH)$. On longer standing in vacuum the dichloral peroxide hydrate arises, whose formation can be explained if one assumes that one molecule of the addition compound dissociates into hydrogen peroxide and chloral, and that the latter then combines with one molecule of the still-undecomposed body:

$$CCl_3{\cdot}CHO + H_2O_2 \rightleftharpoons CCl_3{\cdot}CH(OH)(O{\cdot}OH),$$

$$CCl_3{\cdot}CH(OH)(O{\cdot}OH) + CCl_3{\cdot}CHO = CCl_3{\cdot}CH\overset{O\text{—}O}{\underset{OH\ \ HO}{}}CH{\cdot}CCl_3.$$

The same series of decompositions is undergone by the body slowly on contact with water, instantly on treatment with sodium bicarbonate or acetate. In water the substance is as good as insoluble and is thereby precipitated from a concentrated alcoholic solution in six-sided plates. But if one lets the triturated substance stand 2–3 hours, it dissolves completely, and one finds in the solution only chloral hydrate and hydrogen peroxide. If one further extracts the acidified solution in sodium bicarbonate with ether, there remains after evaporation of the solvent chloral hydrate and a trace of the new compound, which evidently owes its formation to a small amount of hydrogen peroxide taken up by the ether. This easy splitting-off of hydrogen peroxide, incidentally, does not stand isolated, since sulfomonoperacid and benzoperacid show the same behavior in alkaline solution.

It is noteworthy that the addition compound appears to possess no tendency to form, by splitting off water, an inner chloral peroxide, $CCl_3{\cdot}CH(O_2)$. In general, the existence, or at least the formation, of such simple peroxides under the experimental conditions we maintained has become extremely doubtful to us. We must, namely, retract as erroneous what we[4] said about the simple peroxides of the ketones. The pungent-smelling, potassium-iodide-browning, and exploding substances that one obtains on treatment of diethyl and dipropyl ketone with Caro's re-

agent, as well as on decomposition of diacetone diperoxide with concentrated sulfuric acid, are evidently only peracids, formed by splitting of the ketones, or else formed by oxidation of the acetic acid added as solvent, which is readily converted by Caro's reagent into acetoperacid. We were led into this error by the circumstance that at that time we did not yet know the properties of the peracids, and in particular did not know that these are not taken up by sodium carbonate, but are by caustic soda. When we now shook the supposed peroxides with caustic soda, the odor vanished instantly and the solution showed all the properties belonging to the peracids.

Peculiar is the behavior of the dichloral peroxide hydrate toward potassium iodide solution. If one pours the dry substance over with the latter, oxygen is evolved tempestuously with separation of a little iodine.

Peroxides of Acetaldehyde

1 g pure 10% hydrogen peroxide was mixed with 2 g sulfuric acid hydrate and, at 0°, 0.1 g acetaldehyde was added. On standing in the cold, an oily compound soon separates out; if one now warms, until vapors of aldehyde evolve copiously, and then cools rapidly with ice, the oil transforms into a crystal mass which is very volatile, melts below 100°, and explodes very violently on striking or heating. We are inclined to assume that the oily substance corresponds to the above chloral compound and is a **diacetaldehyde peroxide hydrate**, and that the crystals are a **diacetaldehyde diperoxide**, but have not pursued their investigation further on account of their unpleasant properties.

Peroxides of Benzaldehyde.
Dibenzal Diperoxide.

50 g hydrogen peroxide of 2.6 pct. were mixed with 50 g alcohol and 200 g sulfuric acid hydrate, and, under cooling with ice water, 7.5 g benzaldehyde were added all at once. The liquid becomes milky-turbid on shaking, but after some time becomes clear again with separation of a crystal powder. After filtering off, the substance was purified by repeated washing with alcohol and precipitation of the chloroform solution with methyl alcohol. The yield amounted to 40 pct. of the benzaldehyde used. The substance is fairly readily soluble in warm chloroform, ethyl acetate, glacial acetic acid, etc., and separates out on cooling in prisms or needles. It is very sparingly soluble in alcohol and ligroin, somewhat more easily in ether, but insoluble in water. The melting point lies at 202° with gas evolution. The analysis led to the formula of a dibenzal diperoxide, as did the molecular-weight determination given further below.

0.2091 g substance: 0.5255 g CO_2, 0.0931 g H_2O.

$C_{14}H_{12}O_4$. Calc. C 68.85, H 4.92.
Found C 68.54, H 4.95.

Nef[5] prepared, by bringing together benzaldehyde with pure hydrogen peroxide, a substance which corresponds completely in composition and properties to the above chloral compound, which he calls diphenylformal hyperoxide hydrate, a name we have simplified to **dibenzal peroxide hydrate**.

The relationships of Nef's substance to ours follow from the following formulas:

$$2\ C_6H_5{\cdot}COH + H_2O_2 = C_6H_5{\cdot}CH\overset{O—O}{\underset{OH\ \ HO}{}}CH{\cdot}C_6H_5$$

(dibenzal peroxide hydrate)

$$C_6H_5{\cdot}CH\overset{O—O}{\underset{OH\ \ HO}{}}CH{\cdot}C_6H_5 + H_2O_2 = C_6H_5{\cdot}CH\overset{O{\cdot}O}{\underset{O{\cdot}O}{}}CH{\cdot}C_6H_5 + 2\ H_2O$$

(dibenzal diperoxide)

Or in empirical formulas:

$$2\ C_6H_5{\cdot}COH + H_2O_2 = C_{14}H_{14}O_4,$$

$$2\ C_6H_5{\cdot}COH + 2\ H_2O_2 = C_{14}H_{12}O_4 + 2\ H_2O.$$

Nef's substance is therefore an intermediate product and will, in all probability, pass over into ours on treatment with hydrogen peroxide and in the presence of sulfuric acid. The former is, like the chloral compound, extremely labile and decomposes with ease into its components; our compound, on the other hand, being cyclic, is just as stable as the analogous diacetone diperoxide.

The Peroxides of Formaldehyde

Nef[6] advanced the view that the substance prepared by Legler[7] through evaporation of the products arising in the slow combustion of ether is an addition compound of formaldehyde and hydrogen peroxide, formed according to the equation:

$$2\ CH_2O + H_2O_2 = CH_2{\overset{O—O}{\underset{OH\ \ HO}{}}}CH_2,$$

which he named formal hyperoxide hydrate (**diformal peroxide hydrate**). We have no reason to doubt the correctness of this assumption, which places the behavior of formaldehyde alongside that of chloral and benzaldehyde, and wish only to report a new method for preparing the compound obtained by Legler through the action of ammonia on his substance, a method which makes the latter very easily accessible.

Hexamethylene Triperoxide Diamine. (Hexaoxymethylenediamine of Legler.)

Legler obtained this body by treating his hexaoxymethylene hyperoxide, which we, following Nef's lead, designate as diformal peroxide hydrate. One obtains it very easily directly from formaldehyde and hydrogen peroxide by the following procedure.

50 g ammonium sulfate are dissolved warm in an equal amount of ordinary, commercial hydrogen peroxide (about 3%), filtered, and treated at 55° with 5 g of 40% formaldehyde. After a few moments a copious amount of a white powder separates out, which after standing half an hour in the cold is filtered off with suction and washed with water. For the analysis, pure hydrogen peroxide containing only a trace of sulfuric acid was used. The figures agree sufficiently with the composition of Legler's substance.

0.2072 g substance: 0.2564 g CO_2, 0.1204 g H_2O. — 0.1713 g substance: 21.1 cc N (18.5°, 714.5 mm).

$C_6H_{12}O_6N_2$. Calc. C 34.61, H 5.77, N 13.46.

Found C 33.75, H 6.46, N 13.38.

The content of active oxygen was determined by letting the substance stand with acidified potassium iodide solution until everything had gone into solution, and titrating the separated iodine. Since the operation lasted about 6 hours, the amount of spontaneously separated iodine was determined simultaneously by a blank test and subtracted from the former.

0.0859 g substance required 25.05 cc of a $^1/_{10}$-*n.* thiosulfate solution, corresponding to 23.33 pct. active oxygen.

From the formula, 23.06 pct. is calculated.

Legler's body arises from the diformal peroxide hydrate and ammonia according to the following equation:

$$3\ CH_2{\overset{O—O}{\underset{OH\ \ HO}{}}}CH_2 + 2\ NH_3 = C_6H_{12}O_6N_2 + 6\ H_2O,$$

in that the 6 hydroxyls leave with the hydrogen of the ammonia as water, exactly as occurs in the formation of hexamethylenetetramine from formaldehyde hydrate and ammonia:

$$6\ CH_2(OH)_2 + 4\ NH_3 = C_6H_{12}N_4 + 12\ H_2O.$$

The composition of hexamethylene triperoxide diamine is therefore illustrated by the following structural diagram:

$$\begin{array}{c} CH_2{\cdot}O{\cdot}O{\cdot}CH_2 \\ N\,CH_2{\cdot}O{\cdot}O{\cdot}CH_2\,N \\ CH_2{\cdot}O{\cdot}O{\cdot}CH_2 \end{array}$$

The substance is sparingly soluble in all solvents, but can be recrystallized from hot glacial acetic acid, benzene, chloroform, and ethyl acetate, and comes out on cooling in rhombic platelets. Legler states that the substance deflagrates vigorously on heating; we found that on heating, on rubbing, and on impact it explodes as violently as diazobenzene nitrate (nitrate of diazobenzene), and must therefore be handled very cautiously.

On boiling with dilute sulfuric acid the substance goes into solution with formation of ammonia, hydrogen peroxide, and formaldehyde.

Molecular-Weight Determinations

These determinations were carried out for dichloral peroxide hydrate, dibenzal diperoxide, and hexamethylene triperoxide in nitrobenzene solution by the freezing-point-depression method. The constant for the nitrobenzene used was established by experiments with pure naphthalene. The results are satisfactory, considering that the substances are very sparingly soluble and that the chloral compound is, moreover, very unstable.

By experiments with naphthalene, the constant 52.6 was determined for the nitrobenzene used.

Substance	Nitrobenzene (g)	Substance (g)	p	c	72.6·p/c	m (calc.)
Dichloral peroxide hydrate, $C_4H_4Cl_6O_4$	23.95	0.2461 0.4446	1.028 1.856	0.243° 0.438°	307.1 307.6	329
Dibenzal diperoxide, $C_{14}H_{12}O_4$	32.02	0.1011 0.2315	0.316 0.723	0.114° 0.245°	201.2 214.2	244
Hexamethylene triperoxide diamine, $C_6H_{12}O_6N_2$	33.06	0.0804	0.243	0.097°	181.9	208

Note: the multiplier in the table header is printed inconsistently in the original (72.6 in the column heading, 52.6 in the preceding text). The cryoscopic constant for nitrobenzene stated in the text is 52.6.

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[1] On Slow Combustion. Ahrens' Collection of Chemical Lectures, vol. 3, 470 (1899).

[2] These *Berichte* 14, 602; 18, 3343; *Ann. d. Chem.* 217, 383.

[3] *Ann. d. Chem.* 298, 292.

[4] These *Berichte* 33, 125.

[5] *Ann. d. Chem.* 298, 292.

[6] *Ann. d. Chem.* 298, 292, 328.

[7] These *Berichte* 14, 602; 18, 3343; *Ann. d. Chem.* 217, 383.

*English translation of: A. Baeyer and V. Villiger, Berichte der Deutschen Chemischen Gesellschaft **33** (1900), No. 408, pp. 2479–2487.*